

Lab 3: A Lagrange Multiplier by Hand

One equality constraint, stationary points, and a contour picture.

Work in a **Jupyter** notebook. Number the exercises. Do notes **3.1** through **3.3** first. You will need `numpy` and `matplotlib`.

The original assignment PDF is [lab-3-assignment.pdf](#).

When you are done: **File** → **Download as** → **HTML**, then upload the HTML on Canvas.

1. One equality, by hand (3 points)

[$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = 5\theta_1^2 + 3\theta_2^2$ $\quad \text{subject to}$ $\quad g(\boldsymbol{\theta}) = 4\theta_1 + 2\theta_2 - 12 = 0$.]

1. Write the Lagrangian $\mathcal{L}(\boldsymbol{\theta}, \lambda) = J(\boldsymbol{\theta}) + \lambda g(\boldsymbol{\theta})$.
 2. Set $\nabla_{\boldsymbol{\theta}} \mathcal{L} = \mathbf{0}$ and $g=0$. Solve for θ_1 , θ_2 , and λ . Show the algebra in the notebook (markdown or comments), then check the numbers in code.
 3. Say what $\boldsymbol{\theta}$ and λ mean for a model whose parameters must satisfy a computational budget written as $g=0$.
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2. Picture (2 points)

Contour plot of J in the (θ_1, θ_2) plane. Draw the line $g=0$. Mark the constrained minimizer. The unconstrained min is the origin; your star should **not** be there.